

Is it ready to apply *Large Language Models* to frontline hydro practice?

Taking the flood forecasting agent as an example

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On behalf of **HydroAgent Lab**

 Hydrology · AI

 Session HS3.5 · Oral

 Tue 05 May · 11:00–11:10 CEST · Room C



HydroAgent Lab



Frontline hydrologists · AI specialists · Early-career researchers · Scientific software developers
Italy · Germany · United States · China

Community-driven · Volunteer-led · Seeking funding support

Motivation

Physical hydrological models are transparent. *The expert layer that wraps them isn't.*

1 Today's Flood Forecasting Practice

Hydrological models provide the baseline workflow.

Forecasters then adjust by hand: weather context, soil moisture, historical analogs.

→ Much of this expertise remains implicit, subjective, and slow to accumulate.

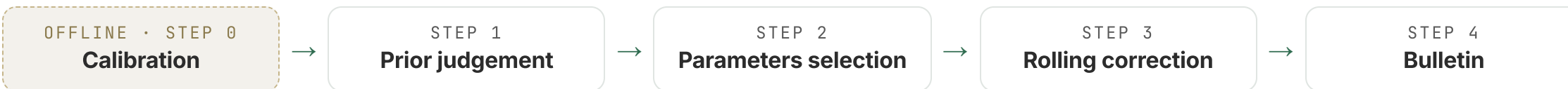
2 Today's AI Practice

ML → black-box rainfall-runoff mappings.

LLMs → mostly chatbots and single-step tool calls.

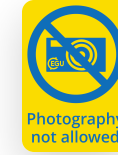
→ The multi-stage, auditable workflow is still missing.

3 Flood forecasting is already a staged process



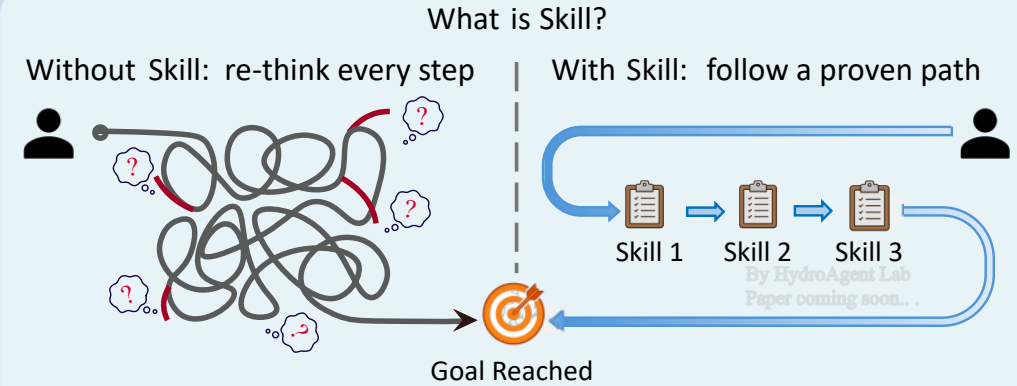
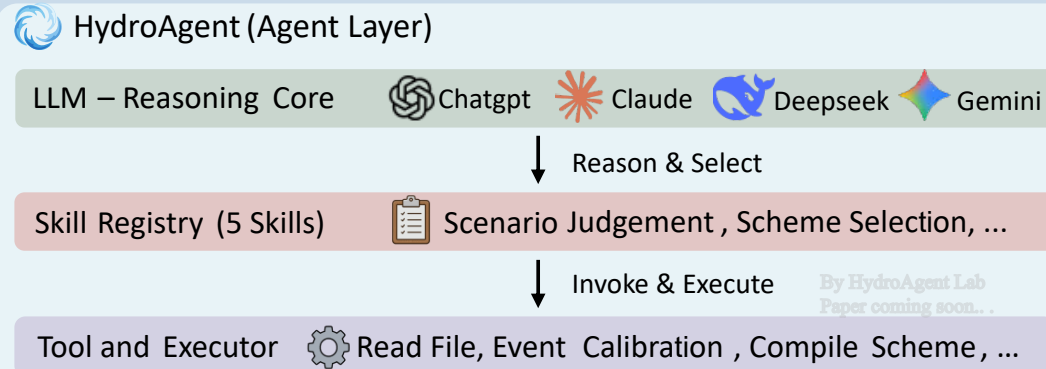
— HydroAgent is our attempt to bring this workflow out of the lab and onto the forecaster's desk. —

HydroAgent: Skill-Driven Architecture

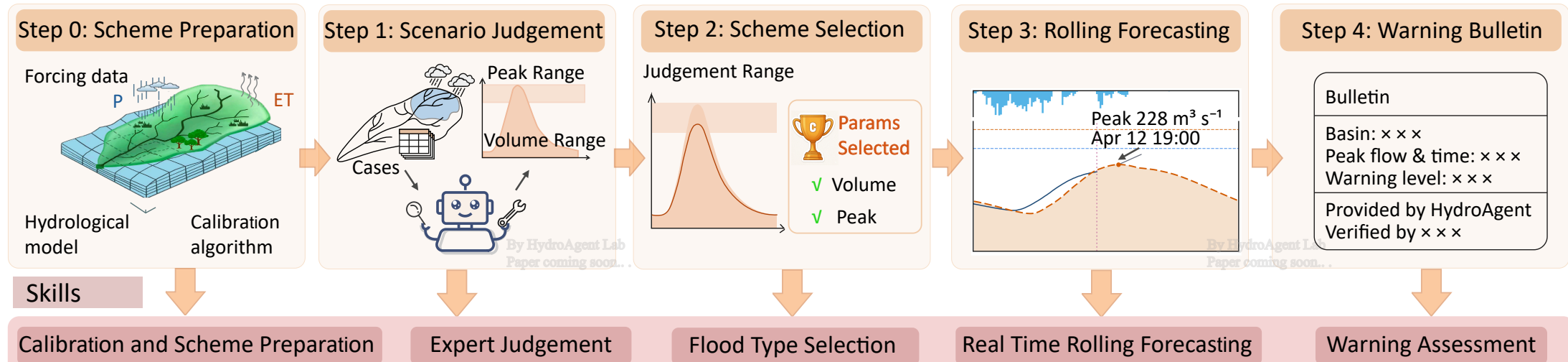


Manuscript under submission;
No photos for this page.

HydroAgent Layered stack



Flood Forecasting Workflow



March 2022, South Yamhill River in the US

Simplified HydroAgent web UI demonstration

US-14194150 basin forecasts a rainfall, ~120 mm rainfall over 87 h (max-24 h 60 mm). 7-day API 22 mm, medium soil moisture. Initial flow 16 m³/s. Please estimate the peak and volume of this flood case.

Step 1 — Scenario Judgment

Parsed scenario → JSON. Retrieved **top-5** flood cases. Reasoning over local rainfall-runoff rules → Class **Type III**. Peak range **210–320 m³/s**. Volume range **0.75–1.30 ×10⁹ m³**. *Paused for Expert Review #1*

Step 2 — Scheme Selection

Selected the optimal parameter set from the Step 0 scheme library, matching the Step 1 peak/volume ranges. Simulated peak **290.86 m³/s**, KGE **0.921**. *Expert Review #2, hydrograph approved.*

Step 3 — Rolling Forecast

Rolling Playback Monitor

running

Basin14194150

Eventevt_20220302T180000

Sim. Time2022-03-01T06:00

Speed1 tick = 2h (real 1s)

Ticks4

Peak Flow275

Peak Time22:00

Warning无

Recent 5 Ticks

03-01T06:00275

03-01T04:00275

Pause

Stop

Current time 2022-03-01T06:00:00

Rainfall

Observed Flow

Forecast Flow

Blue

Yellow

Orange

Red

Flow (m³/s)

1,156

1,000

800

600

400

200

0

02-27 00:00

03-01 00:00

03-03 00:00

03-05 00:00

03-07 00:00

Peak

275 m³/s

2022-03-02 22:00

Step 4 — Bulletin

Peak **290.86 m³/s** @ **2022-03-02 22:00** · RP ≈ 2.1 yr · **No Flood Warning Issued.** *Expert Review #3 signed off.*

BASIN

IDUS-14194150

RiverSouth Yamhill

Area1,358 km²

MODEL & LLM

Hydrological ModelXAJ (14 params)

OptimizerDDS

LLMGPT-5.4

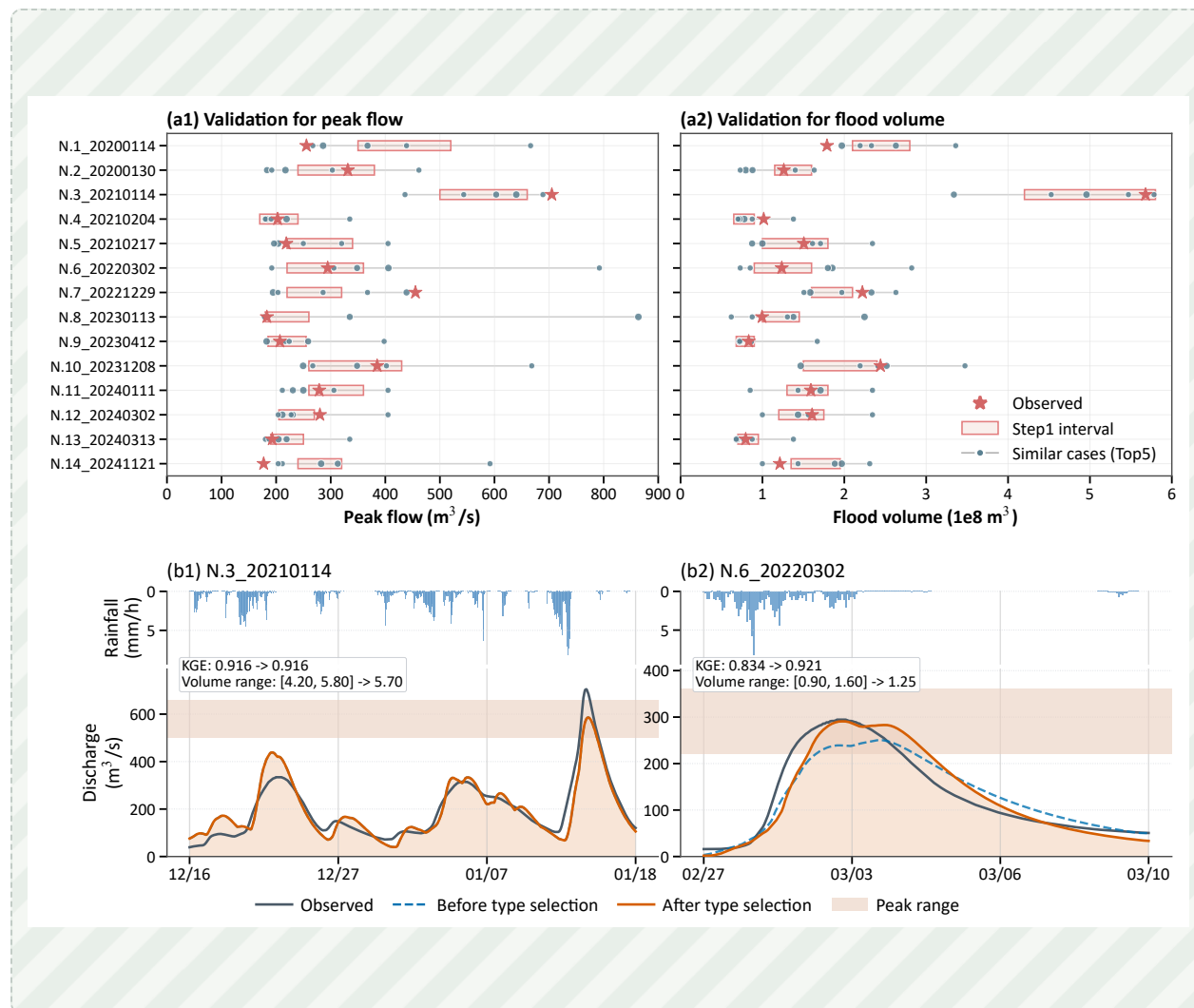
STEP 0 LIBRARY — PER TYPE

Type	Events	KGE
I (peak ≥ 490)	14	0.864
II (350–490)	23	0.942
III (260–350)	22	0.924
IV (205–260)	26	0.896
V (175–205)	28	0.826
Total	113	0.890

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Prior judgment & scheme selection



STEP 1 14 VALIDATION EVENTS · 2020-2024

LLM reasoning narrows the top-5 similar flood cases spread.

Hit rates within a 5% tolerance: **10/14** for peak flow, **11/14** for flood volume. Every judgment range is **narrower than the top-5 spread**. It shows that the LLM is genuinely **refining**, not averaging.

STEP 2 SCHEME SELECTION FROM STEP 0 LIBRARY

Targeted parameter set per event

11/14 events achieve KGE $\geq$ default flood-type classification. Simulated peak falls inside the Step 1 range for **14/14** events, volume for **13/14**.

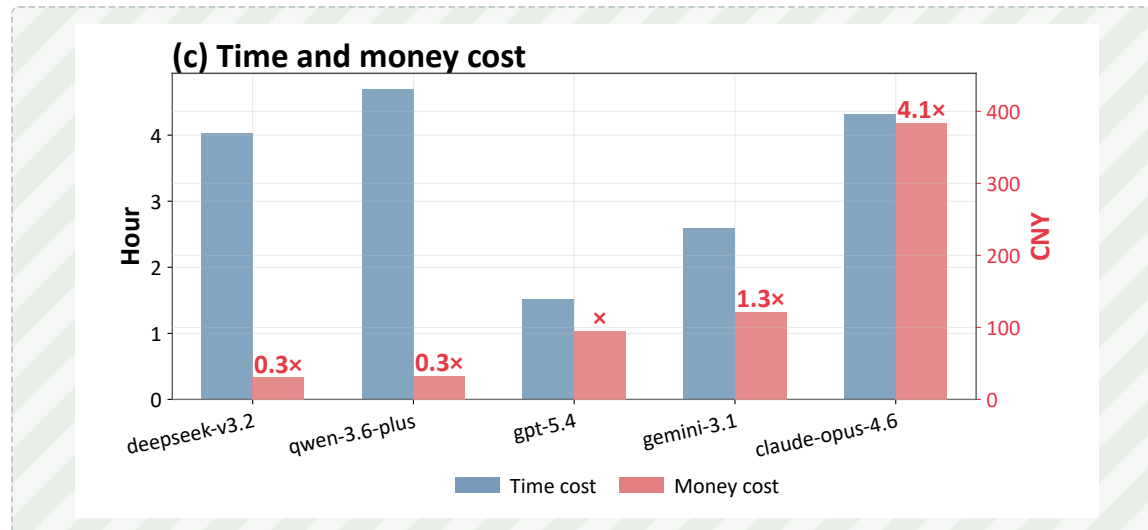
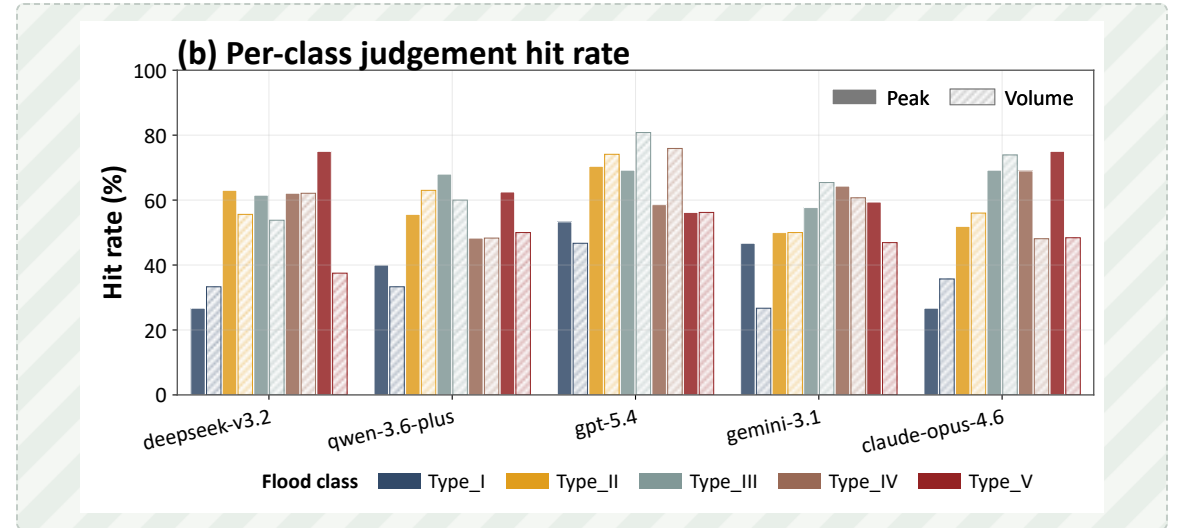
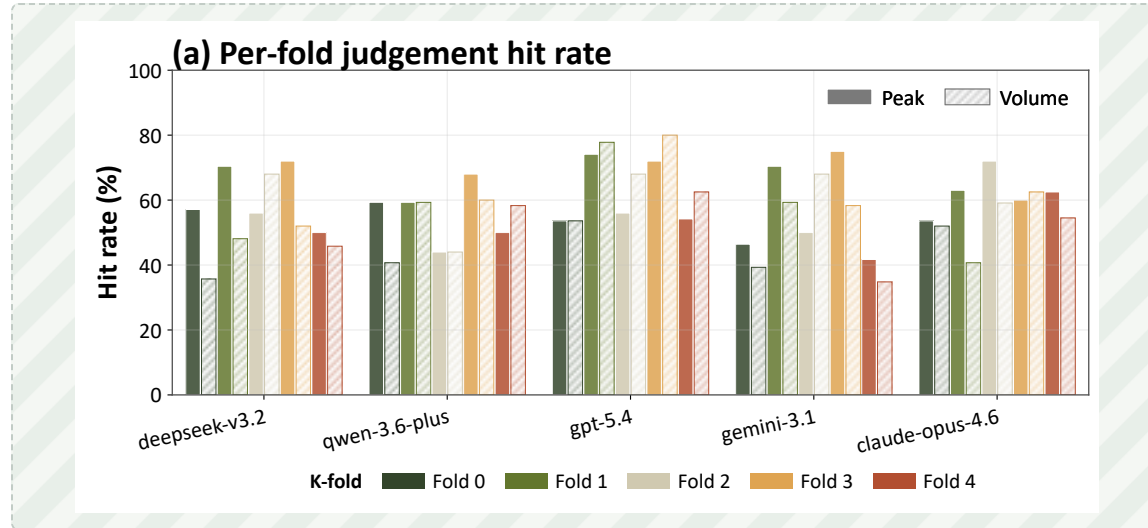
FAILURE MODE

The 3 non-improving events are all **Type V** floods barely above the flood threshold.

Step 1 overestimates, Step 2 inherits the bias.

Step 1 accuracy is the bottleneck, that's why human review checkpoints stay in.

Discussion & Outlook



— 5-FOLD CROSS-VALIDATION * 5 LLMS

- > **No SOTA LLM wins a lot.** Hit rates stay in a 60–85% band across all 5 folds and 5 flood classes. Workflow matters more than the model now.
- > **Cost still varies substantially.** Money spend spans ~13× across LLMS: claude-opus-4.6 costs ~4.1× of gpt-5.4, while deepseek-v3.2 and qwen-3.6-plus are only ~0.3×.
- > **Next: Real-world operational testing.** Deploy HydroAgent during actual flood seasons to accumulate human-in-the-loop feedback.

Three things worth remembering

01

LLMs are not here to replace hydrological models.



They scale expert judgment around those models.

02

The key is not only prompting, it's workflow.



Structured skills make agents reliable, auditable, and reproducible.

03

It's not fully ready yet.



But it's **ready enough to be worth building together.**

We're looking for collaborators, feedback, and real-world testing partners.

Forecasters, modelers, water managers, and AI researchers are all welcome.

Join us.



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contact@hydroagentlab.com

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← scan, valid before May 6
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dby15195876216

WECHAT
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